

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n;
6     scanf("%d", &n);
7
8     int a[100], dp[100];
9     int max = 1;
10
11     for (int i = 0; i < n; i++)
12     {
13         scanf("%d", &a[i]);
14         dp[i] = 1;
15     }
16
17     for (int i = 1; i < n; i++)
18     {
19         for (int j = 0; j < i; j++)
20         {
21             if (a[j] <= a[i] && dp[i] < dp[j] + 1)
22                 dp[i] = dp[j] + 1;
23         }
24
25         if (dp[i] > max)
26             max = dp[i];
27     }
28
29     printf("%d", max);
30
31     return 0;
32 }
33
```

	Input	Expected	Got	
✓	9 -1 3 4 5 2 2 2 2 3	6	6	✓
✓	7 1 2 2 4 5 7 6	6	6	✓

Passed all tests! ✓